Programming Language, Database and Salary Trends: A Developer Survey Analysis

Jismi Therly Jose

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IBM

OUTLINE



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EXECUTIVE SUMMARY



Key findings from the capstone analysis

- Data from a jobs API, web scraping and a developer survey (65,437 respondents) was cleaned and analyzed
- Key findings
  - Median pay rises with age: about $60K at 25-34 to about $110K at 55-64
  - JavaScript, SQL and TypeScript are the most wanted languages; Go and Rust grow fastest
  - PostgreSQL, MySQL and SQLite are the most used databases
- About three quarters of respondents are professional developers, mostly in the US and Western Europe
- A three tab dashboard presents current usage, future trends and demographics

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INTRODUCTION

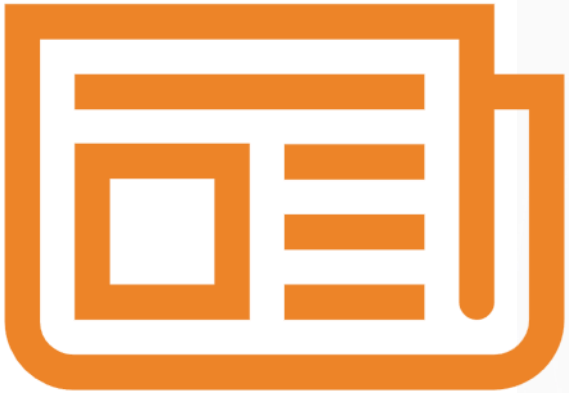


Purpose of the report, target audience and value of the analysis

- Purpose: show which programming languages and databases are used today and which developers want to use next
- Audience: hiring managers, training providers and learners planning their next skills
- Value: evidence to guide hiring, curriculum and learning decisions
- Scope
  - Job postings and popular-language data collected in Module 1
  - Developer survey with 65,437 rows and 114 columns

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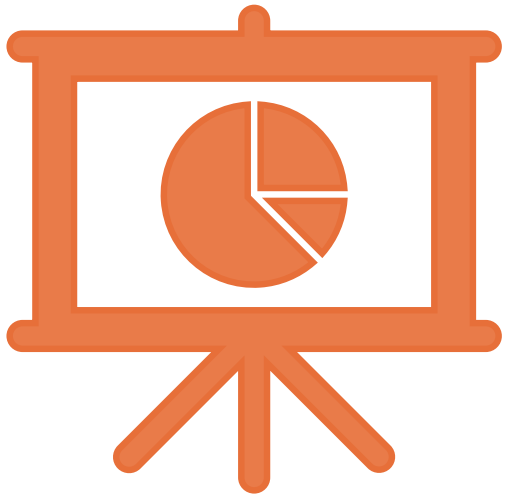
METHODOLOGY



- Data sources: Jobs API (job-postings.xlsx), scraped language table (popular-languages.csv) and the developer survey (survey-data.csv)
- Collection: API calls in a notebook, web scraping with BeautifulSoup, and the survey downloaded into pandas
- Data wrangling
  - Removed 20 duplicate rows and imputed missing values (most frequent value or median)
  - Standardised Country and EdLevel; converted age ranges to numeric midpoints
- Analysis: outliers by the IQR method (978 removed), correlations, then charts and a dashboard
- Dashboard: cleaned subset of 18,846 respondents; technology lists split into counts

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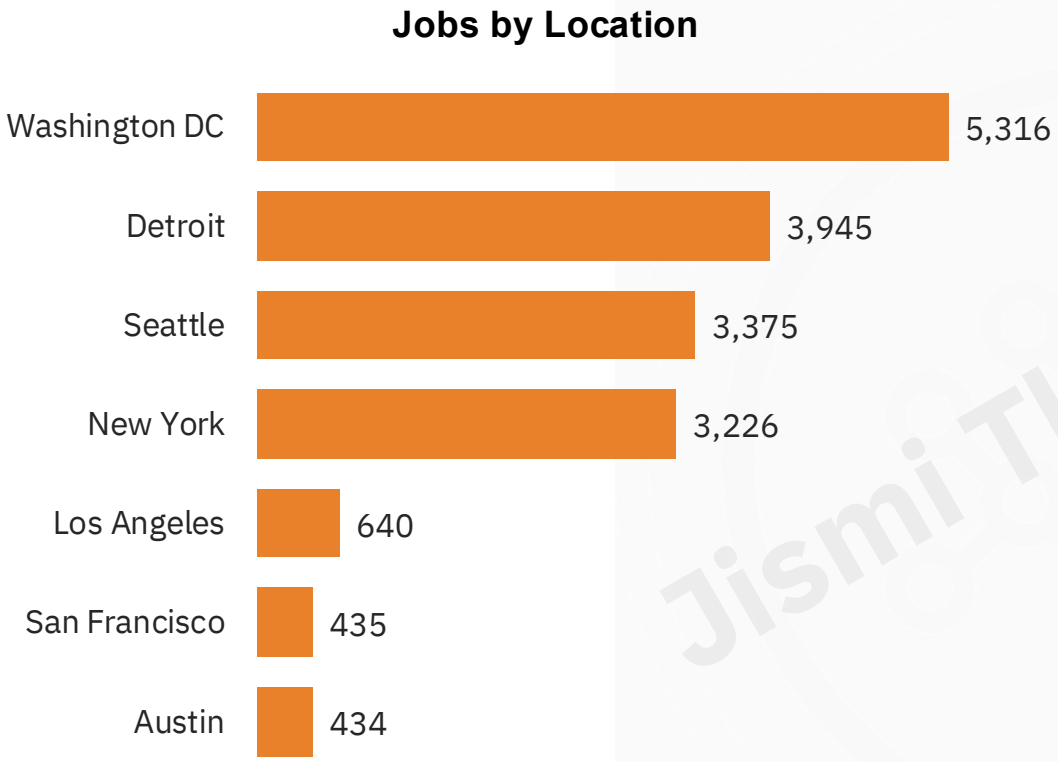
Results

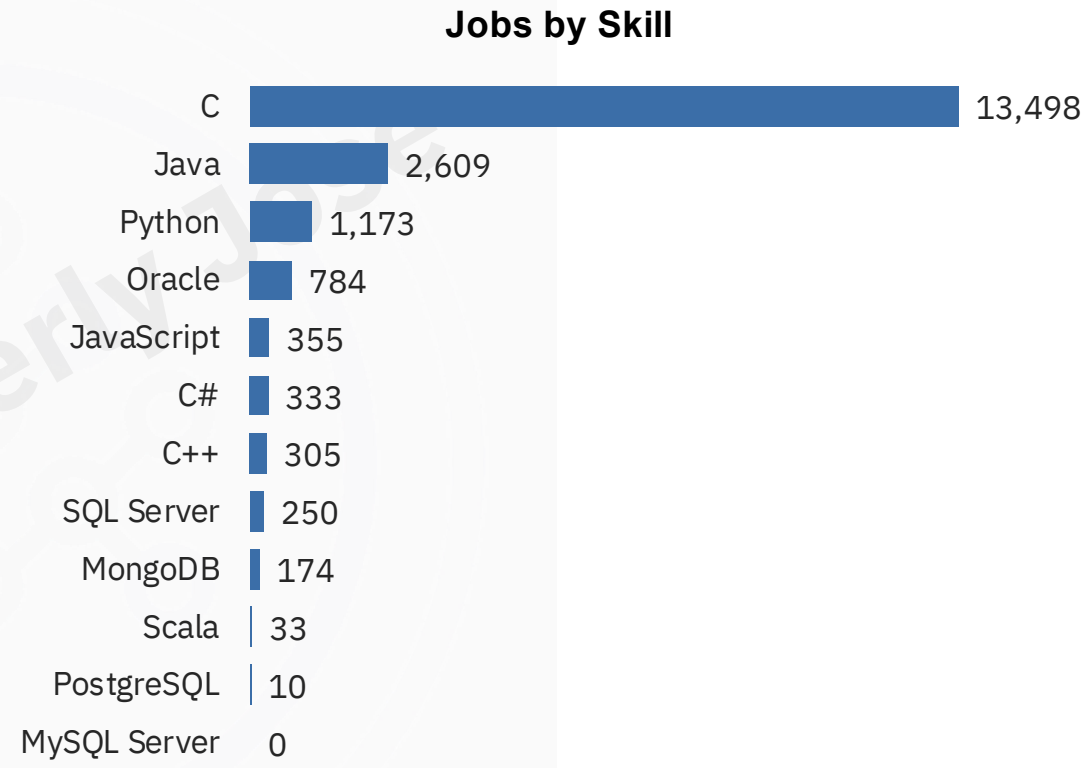


Job postings: demand by location and by key skill
Popular languages: average annual salary by programming language
Language and database trends: what developers use today vs. what they want to use next year

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JOB POSTINGS

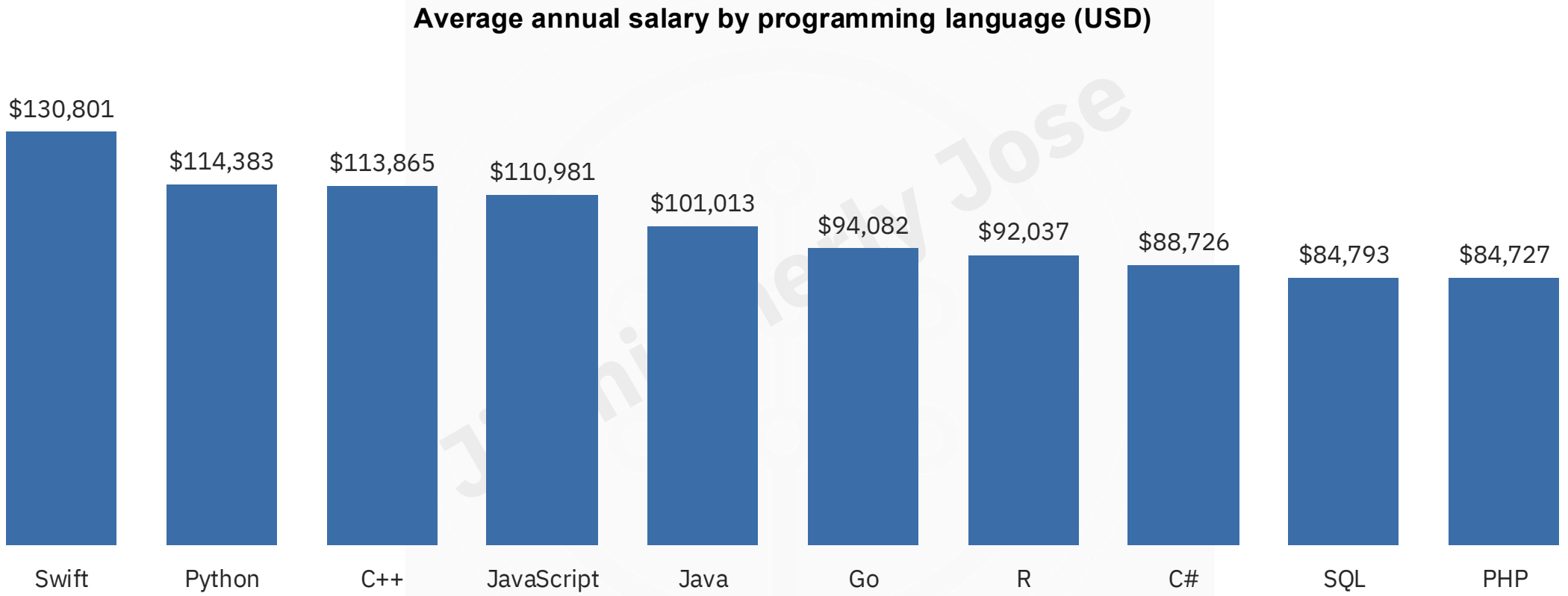




Job postings, highest first. Washington DC (5,316) and Detroit (3,945) have the most postings. Java (2,609) and Python (1,173) lead the named skills. Note: the count for "C" includes any skill text containing "C" (e.g. C#, C++); MySQL Server returned 0 matches.

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POPULAR LANGUAGES

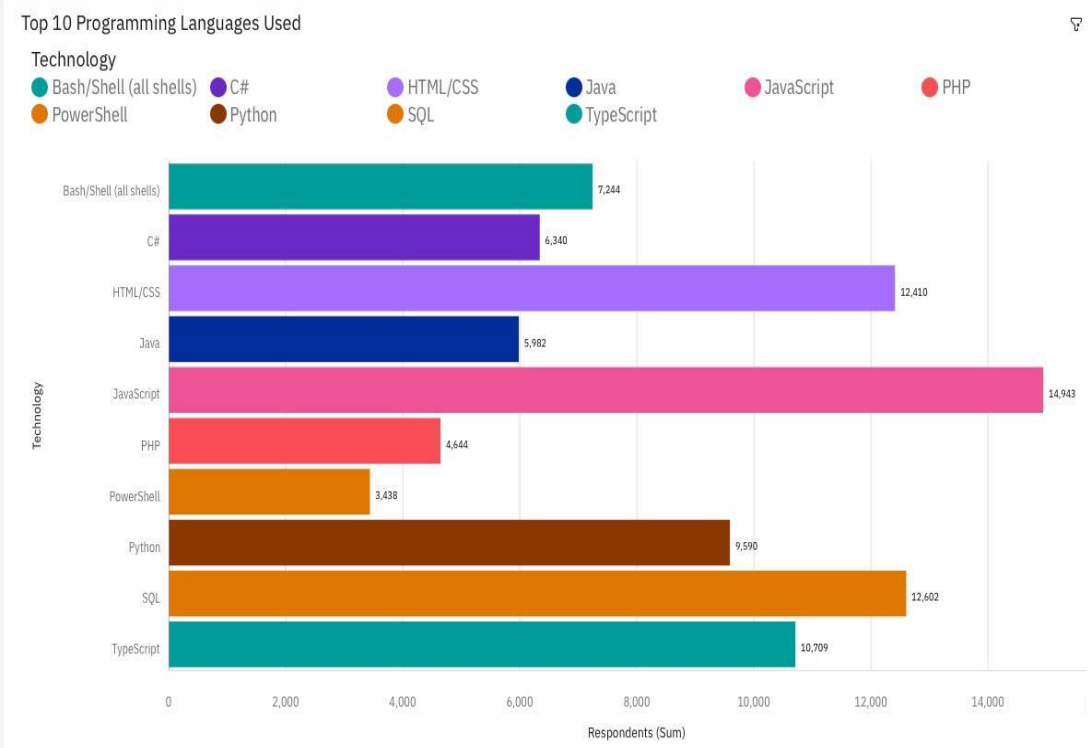


Highest paid first. Swift pays the most ($130,801). Python is second ($114,383) and, as the trend slides show, is also a top 10 language in use and in interest.

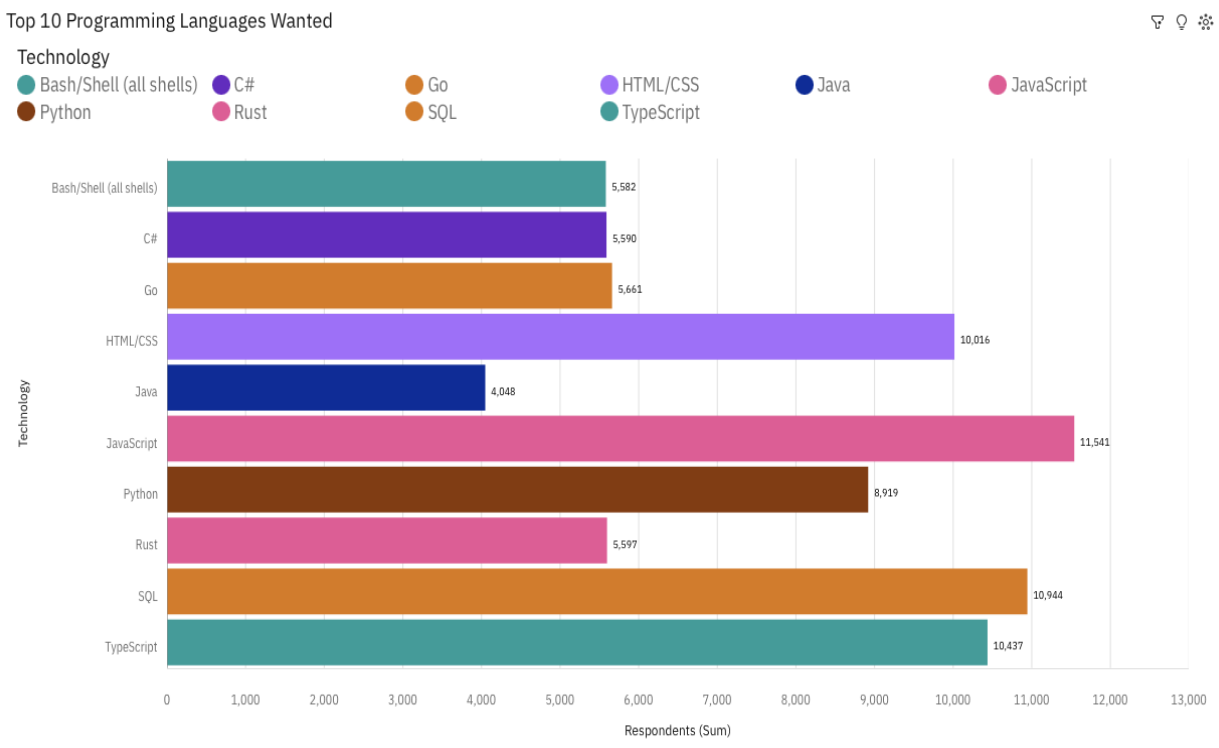
PROGRAMMING LANGUAGE TRENDS

JavaScript, SQL and HTML/CSS lead both years; Go and Rust enter the top 10 next year, while PHP and PowerShell drop out.

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- The core languages stay on top. JavaScript (14,943 used, 11,541 wanted), SQL (12,602 and 10,944) and HTML/CSS (12,410 and 10,016) are in the top four of both charts.
- TypeScript gains ground. It is 4th in use (10,709) but ahead of HTML/CSS in interest (10,437 vs 10,016), which makes it 3rd among wanted languages.
- Python is steady. It is 5th in both charts (9,590 used, 8,919 wanted).
- Go and Rust are the biggest risers. Go goes from 3,305 used to 5,661 wanted (about 1.7×), and Rust from 2,284 to 5,597 (about 2.5×). Both enter the top 10 for next year.
- PHP and PowerShell drop out of the top 10 wanted. Java falls from 5,982 used to 4,048 wanted, about 32% lower.

Implications

- Prioritise SQL and Python. They are the most relevant languages for data work, and demand for both stays strong.
- JavaScript and TypeScript remain the web foundation. TypeScript is the stronger bet for new projects.
- Treat Go and Rust as emerging skills. Interest is growing quickly, but the number currently using them is still small.
- Expect interest in PHP, PowerShell and Java to cool. Existing systems still need support, but new learning is shifting elsewhere.

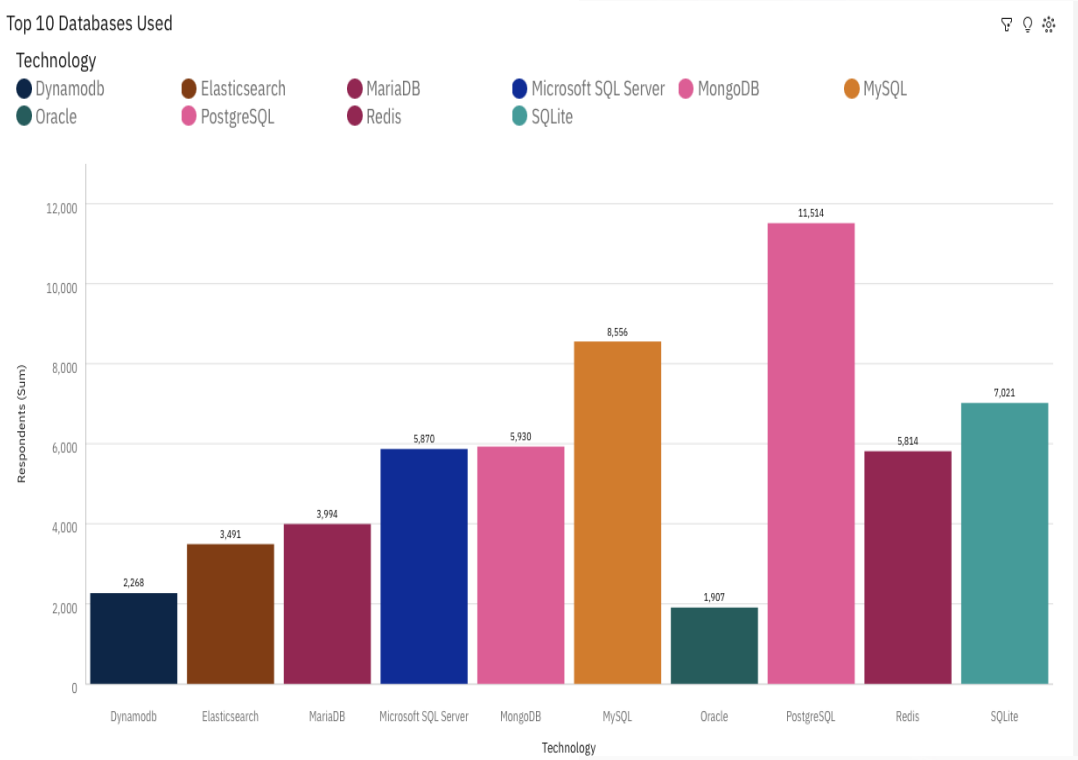
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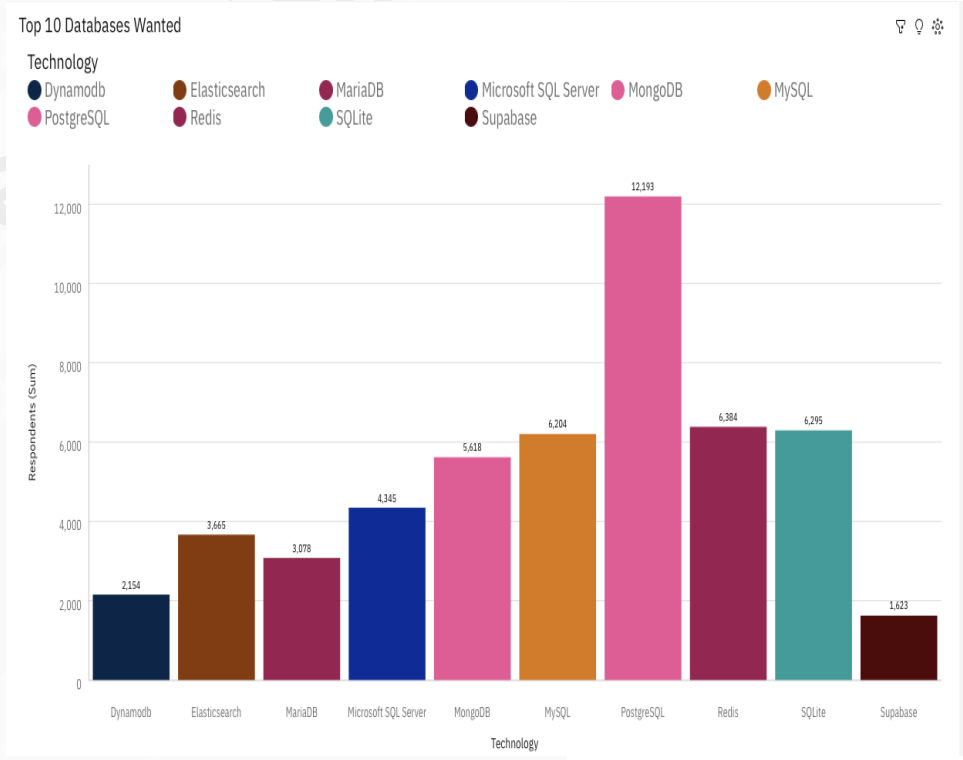
DATABASE TRENDS

PostgreSQL leads both current use and future interest, while MySQL, SQL Server and Oracle lose ground to Redis and the newcomer Supabase.

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL is #1 in both charts. It has 11,514 respondents using it and 12,193 wanting it, so interest is above current use, which is rare in these charts.
- Redis rises from 6th to 2nd. It goes from 5,814 used to 6,384 wanted, ahead of SQLite (6,295) and MySQL (6,204).
- MySQL falls from 2nd to 4th. It drops from 8,556 used to 6,204 wanted, about 27% lower.
- Other traditional databases lose interest. Microsoft SQL Server falls from 5,870 to 4,345 (about 26% lower) and MariaDB from 3,994 to 3,078 (about 23% lower). Oracle (1,907 used) does not appear in the wanted top 10.
- Supabase is a new entry at 1,623 wanted. MongoDB, Elasticsearch and DynamoDB stay roughly stable.

Implications

- Learn PostgreSQL first. It is the strongest choice for both current jobs and future growth.
- Add Redis for caching and speed. Interest is rising there.
- MySQL, SQL Server and Oracle will stay in use but attract less new learning. Existing systems still need support, and new projects are moving toward PostgreSQL.
- Supabase is worth watching. It is a PostgreSQL-based platform, which fits the same trend toward PostgreSQL.

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SALARY VS DEMAND

Language	Avg salary	Used	Wanted	Wanted vs used
Python	$114,383	9,590	8,919	–7%
JavaScript	$110,981	14,943	11,541	–23%
Java	$101,013	5,982	4,048	–32%
Go	$94,082	3,305	5,661	+71%
C#	$88,726	6,340	5,590	–12%
SQL	$84,793	12,602	10,944	–13%

Takeaway: Python pairs high pay with top 5 use and interest. SQL is among the most used and wanted but pays less. Go is the only language here where more people want it than use it.

Salary: web-scraped table. Used and wanted: dashboard.

DASHBOARD



An interactive 3 tab IBM Cognos dashboard summarises the survey: **current technology usage**, **future technology trends** and **respondent demographics**. It compares the technologies developers use today with those they want to use next year.

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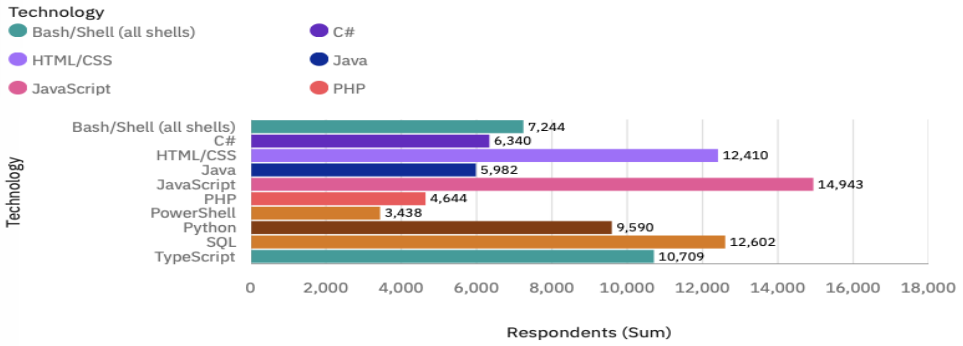
DASHBOARD TAB 1: Current Technology Usage

Technology Usage Count Dashboard

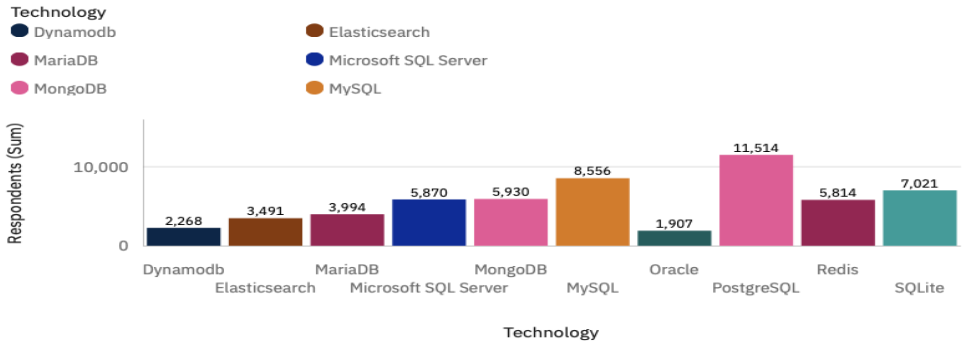
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Current Technology Usage

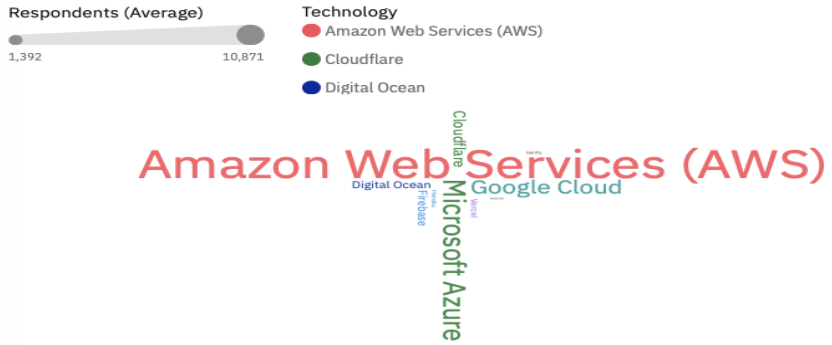
Top 10 Programming Languages Used



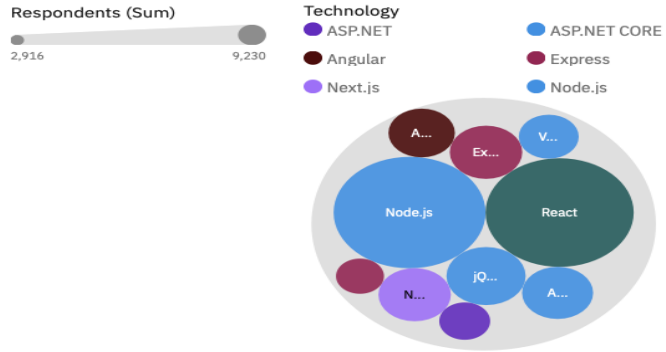
Top 10 Databases Used



Top 10 Cloud Platforms Used



Top 10 Web Frameworks Used



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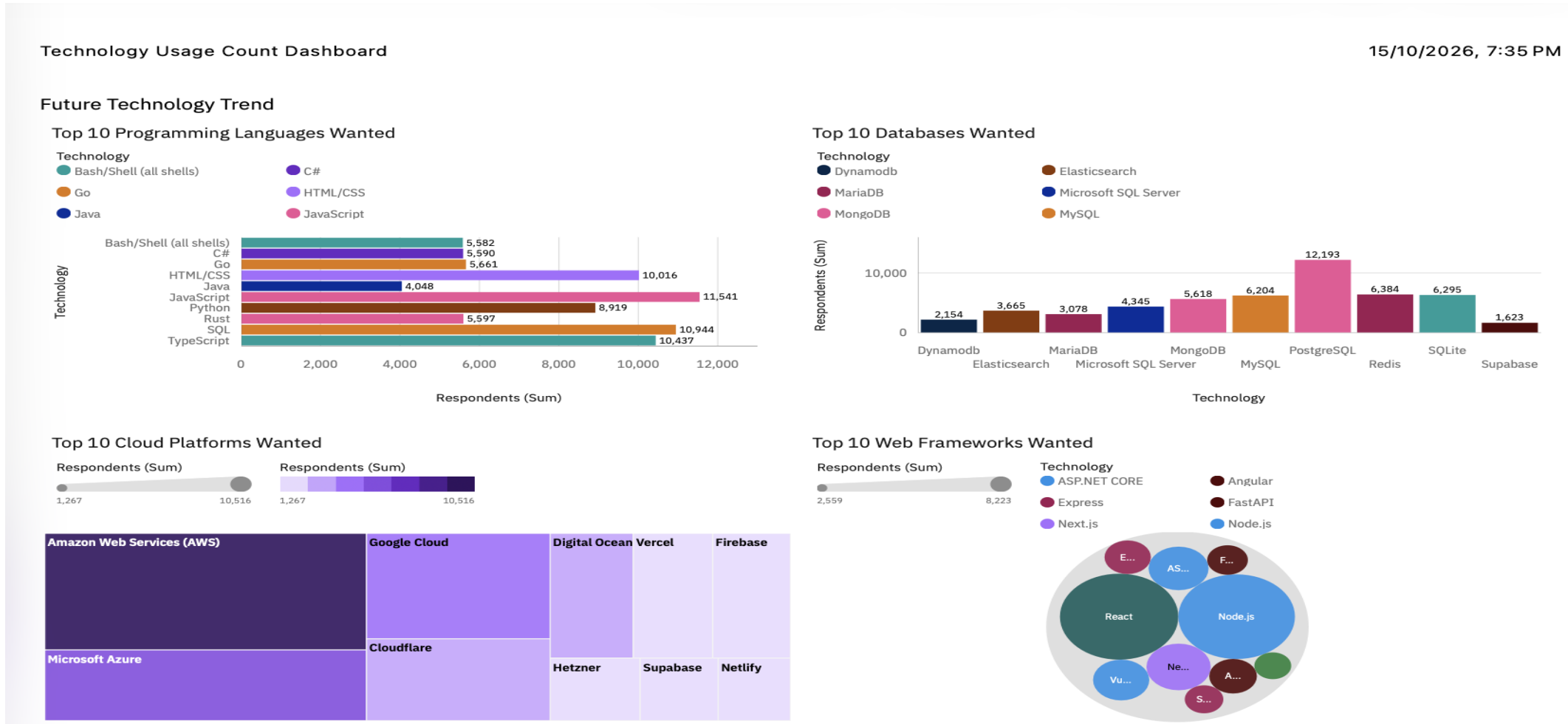
Current Technology Usage – Key Insights

- JavaScript (14,943), SQL (12,602) and HTML/CSS (12,410) are the most used languages. JavaScript is used by about 79% of respondents. Python is 5th at 9,590.
- PostgreSQL is the most used database at 11,514 (about 61%). It is followed by MySQL (8,556) and SQLite (7,021).
- AWS leads cloud platforms at 10,871, ahead of Azure (6,681) and Google Cloud (5,537).
- Node.js (9,230) and React (8,999) are the most used web frameworks.

Today's stack is web centred. JavaScript and SQL are close to universal, PostgreSQL leads databases, and AWS leads the cloud.

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DASHBOARD TAB 2: Future Technology Trends



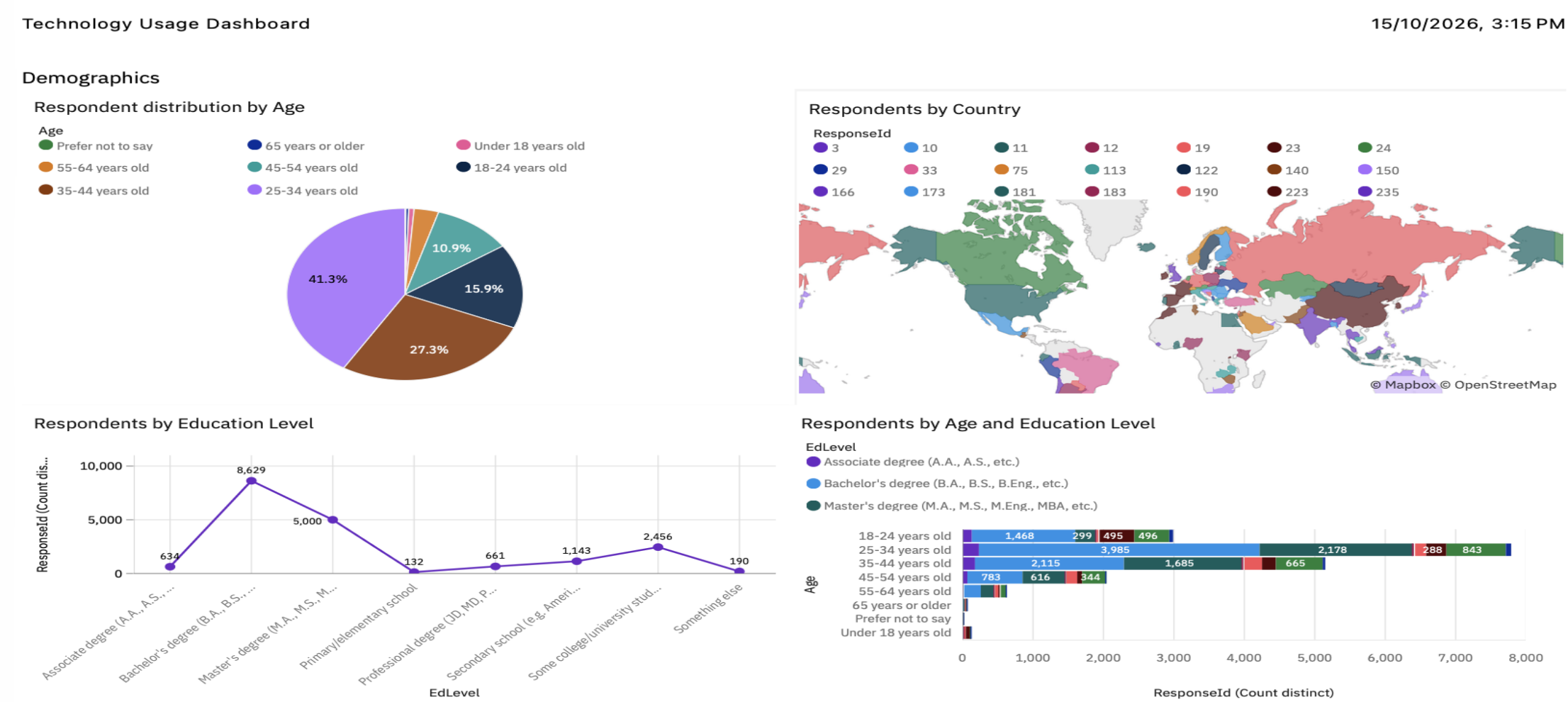
Future Technology Trends – Key Insights

- JavaScript (11,541), SQL (10,944) and TypeScript (10,437) top the languages respondents want to use next year. Go and Rust enter the top 10 (5,661 and 5,597).
- Rust interest is about 2.4× its current use (2,284 → 5,597). Go is about 1.7× (3,305 → 5,661).
- PostgreSQL remains the most-wanted database (12,193) and grows slightly. Redis rises (5,814 → 6,384) and Supabase enters the top 10 (1,623).
- MySQL drops from 8,556 used to 6,204 wanted, about 27% lower.
- Cloud rankings stay the same: AWS (10,516), Azure (6,380), Google Cloud (5,431). React (8,223) overtakes Node.js (8,096), and Next.js (4,458) appears among the most-wanted frameworks.

The core stack stays the same. Growth interest is in Go and Rust, PostgreSQL, Redis and Supabase, and React and Next.js.

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DASHBOARD TAB 3: Demographics



Demographics – Key Insights

- The 25–34 age group is the largest at 41.3% (7,788 respondents). With 35–44 (27.3%), these two groups make up about 69%. 18–24 adds 15.9% and 45–54 adds 10.9%.
- Education is led by a Bachelor’s degree (8,629, about 46%), then a Master’s (5,000, about 27%) and some college (2,456, about 13%). Bachelor’s and Master’s together cover about 72%.
- The map shows respondents from many countries, with the US and Europe most represented.

The respondents are mostly mid career professionals with a university degree. The technology trends above therefore reflect working developers, not students.

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DISCUSSION



- Pay grows with age and experience: median pay rises from $25,000 (18–24) to $109,691 (55–64), with smaller gains at older ages
- Interest is concentrated in a few mainstream technologies: JavaScript, SQL and PostgreSQL lead both use and demand
- Growth is in Go, Rust, Redis and Supabase, in line with the wider industry shift toward cloud-based, high-performance systems
- Pay and demand differ: Swift pays most ($130,801), but Python pairs high pay with top-5 use; SQL is widely used but pays less ($84,793)
- Limits: the sample leans toward the US and Europe, and the dashboard uses a cleaned subset (18,846 of 65,437)

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OVERALL FINDINGS & IMPLICATIONS

Findings

- Pay rises with age: about $60K at 25–34 to $110K at 55–64
- JavaScript, SQL and TypeScript lead wanted languages; Go and Rust show the biggest growth
- PostgreSQL leads databases in use (11,514) and in demand (12,193)
- Washington DC has the most job postings (5,316); Python is the second highest paid language ($114,383)
- Satisfaction shows only a weak link to pay

Implications

- Learn SQL, Python and JavaScript/TypeScript, plus PostgreSQL
- Add one growth skill: Go or Rust, and Redis for speed
- Employers should look beyond pay for satisfaction
- Treat results with care: the sample leans toward the US and Europe

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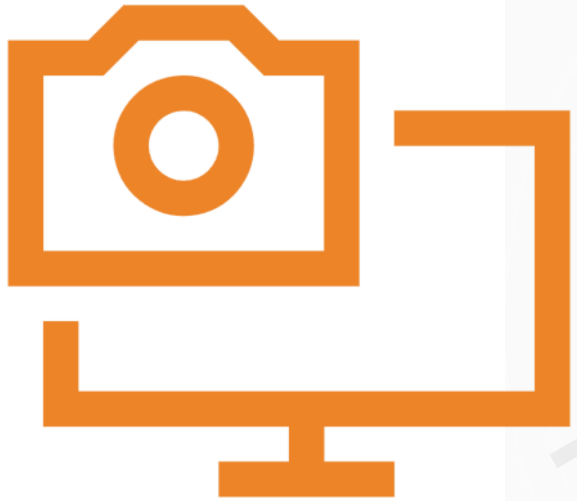
CONCLUSION



- Three data sources were collected, cleaned and visualised: 20 duplicates and 978 outliers were removed
- Experience and age are the clearest drivers of pay (correlation with experience 0.41)
- A few technologies dominate use and interest: JavaScript, SQL, PostgreSQL and AWS
- Go, Rust, Redis and Supabase are the rising technologies to watch
- Recommendation: build a core of SQL, Python and JavaScript/TypeScript, then add one growth skill

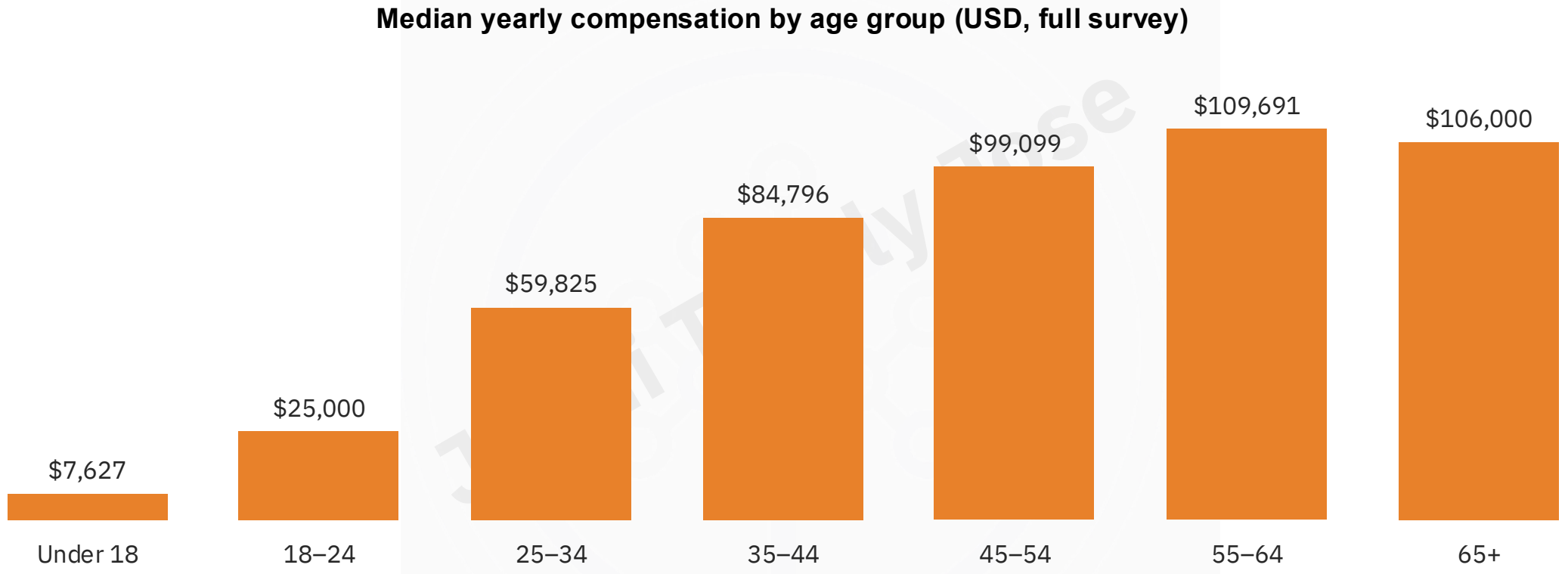
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APPENDIX – DATA SUMMARY



Item	Detail
Developer survey	65,437 rows × 114 columns (20 duplicates removed)
Outliers (IQR method)	978 removed from compensation analysis
Dashboard file	18,846 respondents (cleaned subset)
Job postings (API)	7 locations and 12 key skills
Popular languages	10 languages (web-scraped salaries)
Tools	Python, IBM Cognos Analytics, Excel

APPENDIX – PAY BY AGE



Pay rises with age. Pay rises from $25,000 (18–24) to $109,691 (55–64), then levels off. Compensation and work experience are moderately correlated (r = 0.41); compensation and job satisfaction are only weakly related (about −0.06 to −0.09).